

1. The Assignment — Bus Ticketing System with AI-Powered Search & Booking

Build a web application for a bus ticketing platform with two user roles: an Admin who manages bus routes, schedules, and seat availability, and a Customer who searches for buses, books tickets, and views their bookings.

The AI component: the customer can search for buses using natural language — for example, “I need a bus from Hyderabad to Bangalore tomorrow morning, preferably AC” — and the system interprets the query and returns matching available buses with relevant details.

2. Required Features

- Admin role: add and manage buses with route (origin, destination), departure time, bus type (AC/Non-AC/Sleeper), total seats, price, and availability status
- Customer role: search for buses, view available options with seat count and pricing, book a ticket, and view booking history
- AI-powered natural language search — customer describes their travel need and the system interprets it to return matching available buses ranked by relevance
- Booking flow: customer selects a bus, enters passenger details, confirms booking — seat availability must decrease on booking
- Booking status: Confirmed → Cancelled (customer can cancel a booking, seats are released back)
- Admin dashboard showing: total bookings today, revenue, buses by occupancy rate, route-wise demand
- Prevent overbooking — a bus with zero available seats must not accept new bookings
- Backend API built with Python (Flask or FastAPI) — RESTful endpoints with proper status codes and error handling
- Frontend built with React — separate views for Admin and Customer, clean and functional UI